Engineering Statistics Course Projects

Statistical Inference, Estimation Theory, Reliability and Regression in R

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Summary. A two-phase applied statistics project implemented entirely in **R** (`tidyverse`, `ggplot2`, `MASS`, `pwr`). Phase 1 covers exploratory analysis, assumption checking and hypothesis testing on real datasets, acceptance sampling, simulation and power analysis. Phase 2 covers the theory and practice of parameter estimation, confidence intervals, reliability, and simple linear regression with full diagnostics. Full reports and code are available on GitHub.

Datasets	Sleep & lifestyle data ($n = 374$); student exam scores ($n = 1000$); component lifetimes ($n = 60$); process temperature vs. hardness ($n = 40$)
Methods	Shapiro–Wilk, F-test, Wilcoxon/Mann–Whitney, Welch t-test, Pearson/Spearman, power analysis, Box–Cox, MoM, MLE, Cramér–Rao bound, bootstrap, OLS
Tools	R, ggplot2, dplyr, MASS, pwr, Monte-Carlo simulation

1 Phase 1: Data Analysis and Hypothesis Testing

1.1 Exploratory Visualization of Sleep and Lifestyle Data

Four `ggplot2` figures were used to study relationships between sleep, stress, physical activity and BMI:

- **Sleep duration by disorder (box plot):** median sleep was highest without a disorder (≈ 7.4 h), lower and tightly clustered with insomnia (≈ 6.5 h), and most dispersed with sleep apnea.
- **Stress vs. sleep (jittered scatter plot):** a strong negative, stair-step pattern; sleep fell from above 8 h at stress level 3 to about 6 h at level 8.
- **Stress vs. physical activity (scatter with fitted line):** the fitted line was nearly flat, indicating no meaningful linear association.
- **Daily steps by BMI category (box plot):** step counts decreased with higher BMI (obese group median below 4,000 steps); duplicated category labels were detected and flagged as a data-cleaning issue.

1.2 Hypothesis Testing Workflow

For each question: state H_0/H_1, compute descriptive statistics, test normality (Shapiro–Wilk) and equal variances (F-test), then select a valid test. Normality was rejected in all groups, so the Wilcoxon rank-sum (Mann–Whitney) test was used, which compares the distributions of the two groups.

Question (group means)	Test	p-value	Conclusion
Stress level, male vs. female (6.08 vs. 4.68)	Wilcoxon, two-sided	2.3×10^{-16}	Stress differs by gender; higher in men
Daily steps, activity ≥ 60 vs. < 40 (7,769 vs. 5,165)	Wilcoxon, one-sided	$< 2.2 \times 10^{-16}$	Researcher's claim supported
Sleep hours, no disorder vs. disorder (7.40 vs. 6.50)	Wilcoxon, two-sided	3.6×10^{-9}	Disorders associated with shorter sleep
Activity among men, stress $= 3$ vs. ≥ 7 (65.0 vs. 38.1)	Wilcoxon, one-sided	0.0034	Claim supported (small group, $n = 4$)

```
wilcox.test(`Daily Steps` ~ Activity_Group, data = sleep_activity,
            alternative = "greater", exact = FALSE)
```

1.3 Acceptance Sampling and the OC Curve

For a single-sampling plan with $n = 60$ and acceptance number $c = 3$ (lot size 2000, binomial model), the acceptance probability is $P_a(p) = \sum_{k=0}^{3} \binom{60}{k} p^k (1-p)^{60-k}$, computed with `pbinom` and plotted as the operating-characteristic (OC) curve.

Defect rate p	1%	2%	3%	5%	10%	15%	20%
P_a	0.9969	0.9678	0.8943	0.6473	0.1374	0.0148	0.0010

- At $p = 5\%$ the lot is rejected with probability **35.27%**; at $p = 2\%$ it is accepted with probability **96.78%** (producer's risk 3.22%).
- At $p = 10\%$ the acceptance probability drops to about 14%, showing strong consumer protection.

1.4 Normal Distribution, Simulation and Box–Cox Transformation

Heights of youth volleyball players were modeled as $X \sim N(182, 8^2)$.

- **Exact probability:** $P(175 \le X \le 195) = \Phi(1.625) - \Phi(-0.875) = \mathbf{0.7590}$.
- **Monte-Carlo check:** 1,200 simulated heights gave $\bar{x} = 182.41$, $s = 8.06$; the histogram with fitted density and the Q–Q plot showed no departure from normality.
- **Box–Cox:** the maximum-likelihood λ was close to 1, so the transform left the (already normal) data essentially unchanged, while a log transform distorted the tails.

1.5 Exam-Score Data: Welch Test, Power Analysis, Composite Index, Correlation

- **Preparation course and math score:** with $n_1 = 358$, $n_2 = 642$ and unequal variances, a **Welch** t**-test** was used (justified by the Central Limit Theorem despite a formal Shapiro–Wilk rejection). Means were 69.70 vs. 64.08; $t = 5.787$, $p = 1.04 \times 10^{-8}$; the 95% CI for the difference was $(3.71, 7.52)$ points.
- **Power analysis** (`pwr.t.test`): pooled SD $= 14.93$, Cohen's $d = 0.376$. Replicating the study with 85% power requires **186 students per group** at $\alpha = 0.01$ (128 at $\alpha = 0.05$).
- **Composite academic index:** math, reading and writing scores were z-standardized and averaged. Female students scored higher on this index (0.121 vs. -0.130; Welch $p = 2.6 \times 10^{-5}$, Mann–Whitney $p = 8.3 \times 10^{-6}$).
- **Reading vs. writing:** Pearson $r = 0.9546$ (95% CI $[0.9487, 0.9596]$), Spearman $\rho = 0.9490$, both $p < 2.2 \times 10^{-16}$; reading scores account for about 91% of the variance in writing scores.

2 Phase 2: Estimation Theory, Reliability and Regression

2.1 Lifetime Data and Model Choice

Lifetimes of 60 manufactured components (hours) had mean 356.95, median 237.40 and standard deviation 429.88, so $CV = 1.204$. The right-skewed histogram, mean exceeding median, and a CV close to 1 support an **exponential model** $f(x;\theta) = \theta^{-1} e^{-x/\theta}$.

2.2 Point Estimation: Method of Moments vs. MLE

- Derived both estimators analytically: $\hat{\theta}_{\text{MOM}} = \hat{\theta}_{\text{MLE}} = \bar{X} = \mathbf{356.95}\,\text{h}$, and explained why they coincide.
- **Unbiasedness:** $E(\bar{X}) = \theta$; a Monte-Carlo study ($M = 10{,}000$, $n = 60$) gave a mean estimate of 356.98 (bias $+0.03\,\text{h}$).
- **Consistency:** $\text{Var}(\hat{\theta}) = \theta^2/n \to 0$; simulations for $n \in \{10, 50, 200, 1000, 5000\}$ showed the variance falling from about 12,749 to about 25.
- **Cramér–Rao bound:** $I_1(\theta) = 1/\theta^2$ gives CRLB $= \theta^2/n = 2123.56$ versus an empirical variance of 2165.51, i.e. an efficiency of **98.06%**.

2.3 Confidence Intervals and Reliability

95% CI for θ	Interval (h)	Length (h)
Exact, pivot $2n\bar{X}/\theta \sim \chi^2_{2n}$	$[281.41,\ 467.76]$	186.35
Bootstrap percentile ($B = 5000$)	$[256.46,\ 474.99]$	218.53

The exact chi-square interval is shorter and preferable when the exponential model holds; the bootstrap interval is distribution-free but wider at $n = 60$. **Reliability** at $t = 500$ h, via MLE invariance: $\hat{R}(500) = e^{-500/\hat{\theta}} =$ **0.2464**, with bootstrap 95% CI $[14.23\%,\ 34.90\%]$.

2.4 Simple Linear Regression: Process Temperature and Hardness

Vickers hardness (HV) was modeled as a function of process temperature (°C) for 40 parts ($r = 0.9663$).

Fitted model: $\widehat{HV} = 108.602 + 0.7333 \times \text{Temp}$	
Slope: SE 0.0317, 95% CI $[0.669,\ 0.797]$	$t = 23.15$, $p < 10^{-15}$
Intercept: SE 11.68, 95% CI $[84.97,\ 132.24]$	$R^2 = 0.9338$
$s^2 = 546.90$ ($df = 38$)	Residual SE = 23.39 HV ($\approx 6.4\%$ of mean)

- Derived least-squares estimates by hand ($S_{xx} = 545{,}178.8$, $S_{xy} = 399{,}788.9$, $S_{yy} = 313{,}954.4$) and verified them against `lm()`.
- **OLS = MLE** under normal errors; the estimators differ only in the variance term ($\text{SSE}/(n-2) = 546.90$ unbiased vs. $\text{SSE}/n = 519.56$).
- **ANOVA:** SSR = 293,172.1, SSE = 20,782.3, SST = 313,954.4; temperature explains 93.4% of the variability in hardness.
- **Diagnostics:** no heteroscedasticity pattern in residuals-vs-fitted; Shapiro–Wilk on residuals $W = 0.9711$, $p = 0.391$, supporting normal errors.
- **Prediction at 400°C:** mean hardness 401.93 HV, SE 4.03, 95% CI $[393.78,\ 410.08]$.

```r
fit <- lm(Hardness_HV ~ Temp_C, data = process)
summary(fit); confint(fit); shapiro.test(residuals(fit))
predict(fit, data.frame(Temp_C = 400), interval = "confidence")
```

3 Skills Demonstrated

Statistical inference	Parametric and non-parametric testing, assumption checking, power and sample-size analysis, effect sizes
Estimation theory	Method of moments, MLE, bias, consistency, efficiency, Cramér–Rao bound, pivotal and bootstrap intervals
Modeling	Simple linear regression, ANOVA decomposition, residual diagnostics, composite-index construction
Quality & reliability	OC curves for acceptance sampling, exponential lifetime modeling, reliability estimation
Computing	R (ggplot2, dplyr, MASS, pwr), Monte-Carlo simulation, data visualization, technical reporting